

STOCHASTIC METHODS IN WATER RESOURCES

Unit 5: Geostatistics

Lecture 16: Generalities, descriptive spatial statistics

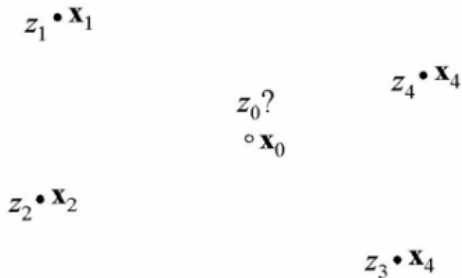
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Generalities

- ▶ **Geostatistics** is the part of statistics that is concerned with **geo-referenced data**, i.e. data that are linked to **spatial coordinates**.
- ▶ To describe the **spatial variation** of the property observed at data locations, the property is modelled with a **spatial random function** (or **random field**) $Z(\mathbf{x})$, $\mathbf{x}^T = (x, y)$ or $\mathbf{x}^T = (x, y, z)$.
- ▶ The focus of geostatistics can be further explained by the figure below.



Suppose that the values of some property (for instance **hydraulic conductivity**) have been observed at the four locations $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$ and $\mathbf{x}_4$ and that, respectively, the values z_1 , z_2 , z_3 and z_4 have been found at these locations.

Generalities

- ▶ **Geostatistics** is concerned with the unknown value z_0 at the non-observed location x_0 . In particular, **geostatistics** deals with:
 1. **spatial interpolation and mapping**: predicting the value of Z_0 at x_0 as accurately as possible, using the values found at the surrounding locations (note that Z_0 is written here in capitals to denote that it is considered to be a **random variable**);
 2. **local uncertainty assessment**: estimating the probability distribution of Z_0 at x_0 given the values found at the surrounding locations, i.e. estimating the **probability density function** $f_z(z_0; x_0 | z_1(x_1), z_2(x_2), z_3(x_3))$. This probability distribution expresses the uncertainty about the actual but unknown value z_0 at x_0 ;
 3. **simulation**: generating realisations of the **conditional random function** $Z(x|z(x_i), i = 1, \dots, 4)$ at many non-observed locations x_i simultaneously (usually on a **lattice** or **grid**) given the values found at the observed locations; e.g. **hydraulic conductivity** is observed at a limited number of locations but must be input to a groundwater model on a grid.
- ▶ **Geostatistics** was first used as a practical solution to estimating ore grades of mining blocks (concentration or percentage of valuable minerals or metals within a block) using observations of ore grades that were sampled preferentially. i.e. along outcrops. Later it was extended to a comprehensive statistical theory for **geo-referenced data**.

Generalities

- ▶ Presently, **geostatistics** is applied in a great number of fields such as **petroleum engineering**, **hydrology**, **soil science**, **environmental pollution** and **fisheries**.
- ▶ Some important **hydrological problems** that have been tackled using **geostatistics** are among others:
 - ▶ spatial interpolation and mapping of rainfall depths and hydraulic heads;
 - ▶ estimation and simulation of representative conductivities of model blocks used in groundwater models;
 - ▶ simulation of subsoil properties such as rock types, texture classes and geological facies;
 - ▶ uncertainty analysis of groundwater flow and -transport through heterogeneous formations (if hydraulic conductivity, dispersivity or chemical properties are spatially varying and largely unknown)

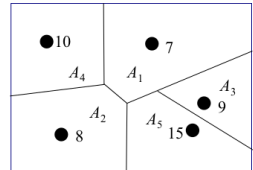
Declustering

- ▶ Looking at figure above it can be seen that not all observation locations are evenly spread in space. Certain location appear to be **clustered**. This can for instance be the case because it is convenient to take a number of samples close together.
- ▶ Another reason could be that certain data clusters are taken purposively, e.g. to estimate the **short distance variance**.
- ▶ If the **histogram** or **cumulative frequency distribution** of the data is calculated with the purpose of estimating the true but unknown **spatial frequency distribution** of an area, it would not be fair to give the same weight to **clustered observations** as to observations that are far from the others. The latter represent a much larger area and thus deserve to be given more weight.
- ▶ To correct for the **clustering effect** **declustering methods** can be used. Here, one particular **declustering method** called **polygon declustering** is illustrated in the figure.
- ▶ The figure shows schematically a spatial array of measurement locations. The objective is to estimate the **spatial statistics** (**mean**, **variance**, **histogram**) of the property (e.g. **hydraulic conductivity**) of the field.

$$w_1 = \frac{A_1}{A_1 + A_2 + A_3 + A_4 + A_5}$$
$$w_2 = \frac{A_2}{A_1 + A_2 + A_3 + A_4 + A_5}$$

etc.

$$A_1 = 0.30$$
$$A_2 = 0.25$$
$$A_3 = 0.10$$
$$A_4 = 0.20$$
$$A_5 = 0.15$$



$$m_z = 0.3 \cdot 7 + 0.25 \cdot 8 + 0.10 \cdot 9 + 0.20 \cdot 10 + 0.15 \cdot 15 = 9.25$$

$$s_z^2 = 0.3 \cdot (-2.25)^2 + 0.25 \cdot (-1.25)^2 + 0.10 \cdot (-0.25)^2 +$$

$$0.20 \cdot (0.75)^2 + 0.15 \cdot (5.75)^2 = 6.99$$

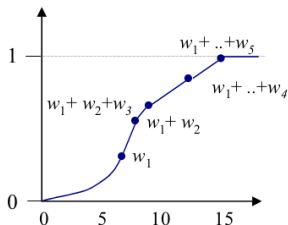
Declustering

- ▶ The idea is to draw **Thiessen polygons** around the observation locations first: by this procedure each location of the field is assigned to the closest observation.
- ▶ The relative sizes of the **Thiessen polygons** are used as **declustering weights**:

$$w_i = \frac{A_i}{\sum_j A_j}.$$
- ▶ Using these weights the **declustered histogram** and **cumulative frequency distribution** can be calculated as shown in the figure below, as well as the **declustered moments** such as the **mean** and **variance**:

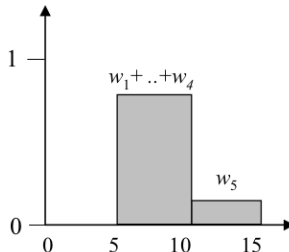
$$m_z = \sum_{i=1}^n w_i z_i \quad (1)$$

Declustered cum. Freq. Distr.



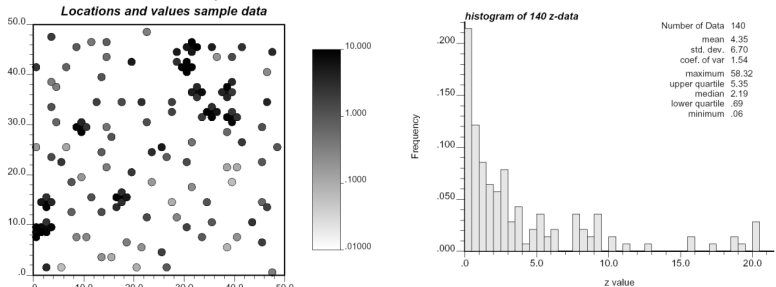
$$s_z^2 = \sum_{i=1}^n w_i (z_i - m_z)^2 \quad (2)$$

Declustered histogram



Declustering

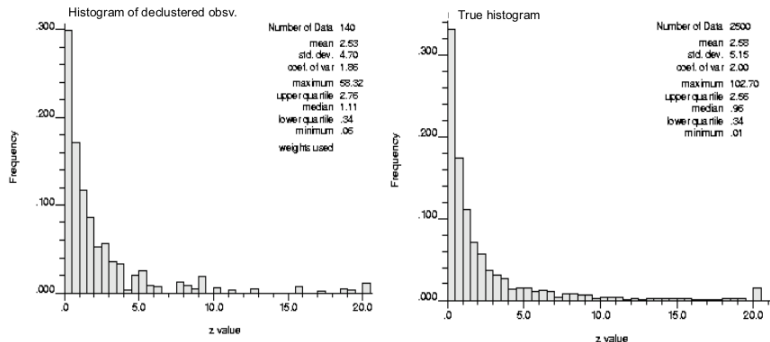
- ▶ The effect of **declustering** can be demonstrated using the synthetic data-set shown in the figure below (all of the larger numerical examples shown in here are based on this data set).



Descriptive spatial statistics

Declustering

- ▶ The figure below shows the **declusterd histogram** based on the 140 data and the **"true" histogram** based on 2500 values. Clearly, they are very much alike, while the histogram based on the non-weighted data (see figure) is much different.

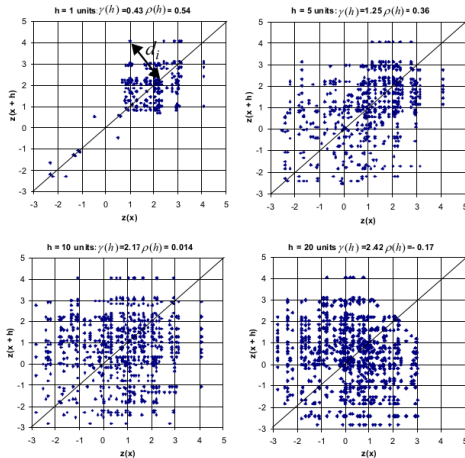


- ▶ The estimated **mean** without weighting equals 4.35 which is much too large. The reason is the existence of clusters with very high data values present in the observations (see figure above).
- ▶ **Declustering** can correct for this as can be seen from the **declustered mean** in the figure above which is 2.53 and very close to the true mean of 2.58.

Descriptive spatial statistics

Semivariance and correlation

- ▶ Using the synthetic data set of 140 observations we will further illustrate the concept of the **semivariogram** and the **correlation function**.
- ▶ Figure below shows scatter plots of the $z(x)$ and $z(x + h)$ for $|h| = 1, 5, 10, 20$ units (pixels) apart. For each pair of points the distance d_i to the one-to-one line it can be calculated.



Descriptive spatial statistics

Semivariance and correlation

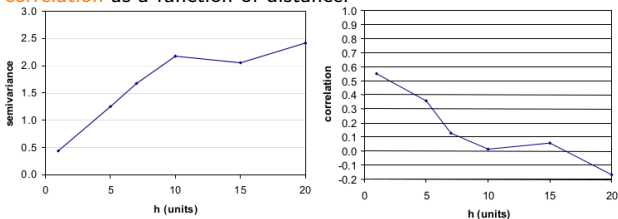
- ▶ The **semivariance** of a given distance is given by (with $n(h)$ the number of pairs of points that are a distance $h = |\mathbf{h}|$ apart)

$$\hat{\gamma}(h) = \frac{1}{2n(h)} \sum_{i=1}^{n(h)} d_i^2 = \frac{1}{2n(h)} \sum_{i=1}^{n(h)} [z(\mathbf{x} + \mathbf{h}) - z(\mathbf{x})]^2 \quad (3)$$

and the **correlation coefficient**

$$\hat{\rho}(h) = \frac{1}{n(h)} \sum_{i=1}^{n(h)} \frac{z(\mathbf{x} + \mathbf{h})z(\mathbf{x}) - m_{z(\mathbf{x}+\mathbf{h})}m_{z(\mathbf{x})}}{S_{z(\mathbf{x}+\mathbf{h})}S_{z(\mathbf{x})}} \quad (4)$$

$m_{z(\mathbf{x}+\mathbf{h})}$, $m_{z(\mathbf{x})}$ and $S_{z(\mathbf{x}+\mathbf{h})}$, $S_{z(\mathbf{x})}$ are the **mean** and **variance** of the $z(\mathbf{x} + \mathbf{h})$ and $z(\mathbf{x})$ data values respectively. The figure below shows plots of the **semivariance** and the **correlation** as a function of distance.



- ▶ These plots are called the **semivariogram** and the **correlogram** respectively. If we imagine the data z to be observations from a realisation of a random function $Z(\mathbf{x})$ and this random function is assumed to be intrinsic or wide sense **stationary** then equations 3 and 4 are estimators for the **semivariance function** and the **correlation function**.

Spatial interpolation by kriging

Generalities

- ▶ **Kriging** is a collection of methods that can be used for **spatial interpolation**.
- ▶ **Kriging** provides **optimal linear predictions** at non-observed locations by assuming that the unknown **spatial variation** of the property is a realisation of a **random function** that has been observed at the data points only.
- ▶ Apart from the prediction, **kriging** also provides the **variance** of the prediction
- ▶ Two **kriging** variants are discussed: **simple kriging** and **ordinary kriging** which are based on slightly different **random function** models.

Spatial interpolation by kriging

Simple kriging

The most elementary of **kriging methods** is called **simple kriging** and is based on a **random function** that is **wide sense stationary**, i.e. with the following properties

$$E[Z(\mathbf{x})] = \mu_Z = \text{constant}$$

$$\text{var}[Z(\mathbf{x})] = E[Z(\mathbf{x}) - \mu_Z]^2 = \sigma_Z^2 = \text{constant (and finite)}$$

$$\text{COV}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)] = E[(Z(\mathbf{x}_1) - \mu_Z)(Z(\mathbf{x}_2) - \mu_Z)] = C_Z(\mathbf{x}_2 - \mathbf{x}_1) = C(\mathbf{h})$$

Simple kriging is the appropriate **kriging method** if the **random function** is **second order stationary** and the **mean** of the **random function** $E[Z(\mathbf{x})] = \mu$ is known without error. With **simple kriging** a predictor $\hat{Z}(\mathbf{x}_0)$ is sought that

1. is a linear function of the surrounding data,
2. is unbiased: $E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = 0$,
3. and has the smallest possible error, i.e. $\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)$ is minimal.

A **linear** and **unbiased** predictor is obtained when considering the following **weighted average** of deviations from the **mean**:

$$\hat{Z}(\mathbf{x}_0) = \mu_Z + \sum_{i=1}^n \lambda_i [Z(\mathbf{x}_i) - \mu_Z] \quad (5)$$

with $Z(\mathbf{x}_i)$ the values of $Z(\mathbf{x})$ at the surrounding observation locations.

Spatial interpolation by kriging

Simple kriging

Usually, not all observed locations are included in the predictor, but only a limited number of locations within a given **search neighbourhood**. Equation 5 is unbiased by definition:

$$E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = \mu_Z + \sum_{i=1}^n \lambda_i [Z(\mathbf{x}_i) - \mu_Z] - E[Z(\mathbf{x}_0)] = \mu_Z + \sum_{i=1}^n \lambda_i [\mu_Z - \mu_Z] - \mu_Z = 0 \quad (6)$$

The **weights** λ_i should be chosen such that the **prediction error** is minimal. However, as the real value $z(\mathbf{x}_0)$ is unknown, we cannot calculate the **prediction error**. Therefore, instead of minimizing the prediction error we must be satisfied with minimizing the **variance** of the prediction error $\text{var}[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)]$. Because the predictor is unbiased, the **variance** of the **prediction error** can be written as

$$\begin{aligned} \text{var}[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] &= E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)]^2 = \\ E \left[\sum_{i=1}^n \lambda_i [Z(\mathbf{x}_i) - \mu_Z] - [Z(\mathbf{x}_0) - \mu_Z] \right]^2 &= \\ \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j E [[Z(\mathbf{x}_i) - \mu_Z][Z(\mathbf{x}_j) - \mu_Z]] - & \quad (7) \\ 2 \sum_{i=1}^n \lambda_i E [[Z(\mathbf{x}_i) - \mu_Z][Z(\mathbf{x}_0) - \mu_Z]] + E[Z(\mathbf{x}_0) - \mu_Z]^2 & \end{aligned}$$

Spatial interpolation by kriging

Simple kriging

Using the definition of the **covariance** of a **second order stationary stochastic field** $E [[Z(\mathbf{x}_i) - \mu_Z][Z(\mathbf{x}_j) - \mu_Z]] = C_Z(\mathbf{x}_i - \mathbf{x}_j)$ and $C_Z(0) = \sigma_Z^2$, we obtain for the **variance** of the **prediction error**

$$\text{var}[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) - 2 \sum_{i=1}^n \lambda_i C_Z(\mathbf{x}_i - \mathbf{x}_0) + \sigma_Z^2 \quad (8)$$

To obtain the minimum value of Equation 8 we have to equate all its partial derivatives with respect to the λ_i to zero:

$$\frac{\partial}{\partial \lambda_i} \text{var}[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = 2 \sum_{j=1}^n \lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) - 2 C_Z(\mathbf{x}_i - \mathbf{x}_0) = 0 \quad i = 1, 2, \dots, n \quad (9)$$

This results in the following system of n equations referred to as the **simple kriging system**

$$\sum_{j=1}^n \lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) = C_Z(\mathbf{x}_i - \mathbf{x}_0) \quad i = 1, 2, \dots, n \quad (10)$$

The n unknown values λ_i can be uniquely solved from these n equations if all the $\mathbf{x}_i$ are different. The predictor in equation 5 with the λ_i found from solving equations 10 is the one with the minimum **prediction error variance**. This **variance** can be calculated using equation 8. However, it can be shown that the **variance** of the **prediction error** can be written in a simpler form as

$$\text{var}[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = \sigma_Z^2 - \sum_{i=1}^n \lambda_i C_Z(\mathbf{x}_i - \mathbf{x}_0) \quad (11)$$

Spatial interpolation by kriging

Simple kriging

The **error variance** very nicely shows how **kriging** takes advantage of the spatial dependence of $Z(\mathbf{x}_i)$. If only the **marginal probability distribution** had been estimated from the data and the spatial coordinates had not been taken into account, the best prediction for every non-observed location would have been the **mean** μ_Z .

Consequently, the **variance** of the **prediction error** would have been equal to σ_Z^2 . As the larger **kriging weights** are positive, it can be seen from equation 11 that the **prediction error variance** of the **kriging predictor** is always smaller than the variance of the **random function**.

To obtain a **positive error variance** using equation 11 the function $C(\mathbf{h})$ must be **positive definite**. This means that for all possible $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^N$ ($N = 1, 2$ or 3) and for all $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$ the following inequality must hold

$$\sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) > 0 \quad (12)$$

- ▶ It is difficult to ensure that this is the case for any **covariance function**. Therefore, we cannot just estimate a **covariance function** directly from the data for a limited number of separation distances and then obtain a **continuous function** by **linear interpolation** between the experimental points (such as in figure above).
- ▶ If such a **covariance function** were used in equation 10, equation 11 would not necessarily lead to a positive estimate of the **prediction error variance**.

Spatial interpolation by kriging

Simple kriging

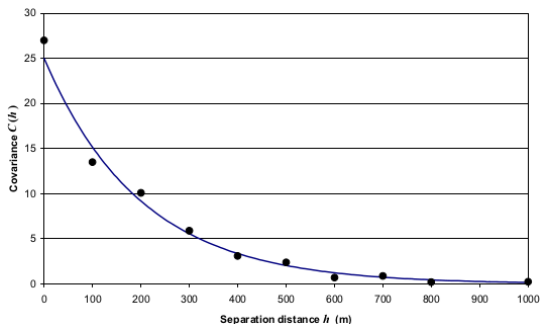
- ▶ In fact, there are only a limited number of functions for which it is proven that inequality 12 will always hold. So the practical solution used in **kriging** is to take one of these 'permissible' functions and fit it through the points of the experimental **covariance function**.
- ▶ Next, the values of the fitted function are used to build the **kriging system** (see equation 10 and to estimate the kriging variance using equation 11.
- ▶ The table below gives a number of **covariance functions** that can be used for simple **kriging** (i.e. using a wide sense stationary random functions). Of course, in case of second order stationarity the parameter c should be equal to the **variance** of the random function: $c = \sigma_Z^2$.

Exponential covariance	$C_Z(h) = ce^{-h/a}$ for $h \geq 0, a > 0$
Gaussian covariance	$C_Z(h) = ce^{-h^2/a^2}$ for $h \geq 0, a > 0$
Spherical covariance	$C_Z(h) = \begin{cases} c \left[1 - \frac{3}{2} \left(\frac{h}{a} \right) + \frac{1}{2} \left(\frac{h}{a} \right)^3 \right], & \text{if } h < a \\ 0, & \text{if } h \geq a, h \geq 0, a > 0 \end{cases}$
Nugget model	$\rho_Z(h) = \begin{cases} c, & \text{if } h = 0 \\ 0, & \text{if } h > 0 \end{cases}$

Spatial interpolation by kriging

Simple kriging

- The figure below shows an example of an **exponential model** that is fitted to the **estimated covariances**.



- Any linear combination of a **permissible covariance model** is a permissible covariance model itself. Often a combination of a **nugget model** and another model is observed, e.g:

$$C(h) = \begin{cases} c_0 + c_1, & \text{if } h = 0 \\ c_1 e^{-h/a}, & \text{if } h > 0 \end{cases} \quad (13)$$

where $c_0 + c_1 = \sigma_Z^2$. In this case c_0 is often used to model the part of the variance that is attributable to **observation errors** or **spatial variation** that occurs at distances smaller than the minimal distance between observations.

Spatial interpolation by kriging

Simple kriging: practical application

The practical application of **simple kriging** would involve the mapping of some variable observed at a limited number of locations. In practice, the kriging routine would consist of the following steps:

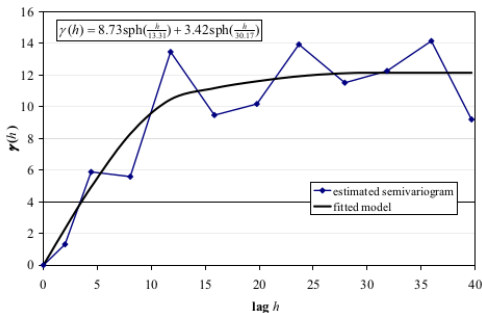
1. Estimate the **mean** and the **covariance** function from the observed data after **declustering**.
2. Fit a permissible **covariance model** to the **experimental semivariogram**. Usually one does not estimate the **covariance function** but the **semivariogram** when **kriging**. The **semivariogram** is somewhat better suited for estimating data that are **irregularly distributed** in space. After fitting a **semivariogram function** that is suited for **wide sense stationary processes**, the **covariance function** can be obtained as:

$$C_Z(\mathbf{h}) = \sigma_Z^2 - \gamma_Z(\mathbf{h}) \quad (14)$$

Spatial interpolation by kriging

Simple kriging: practical application

The **semivariogram** of the synthetic data set based on 140 data points and the fitted model (**spherical model**) is below.



Note that if **kriging** is used for making maps, the locations where the predictions are made are usually located on a grid. So, when in the following steps we refer to a prediction location $\mathbf{x}_0$ we refer to a location on this grid. Thus, the following steps are repeated for every grid node:

3. Solve the **simple kriging system** using equation 11 and the **covariance function** $C_Z(\mathbf{h})$ (see equation 14). λ_i are obtained for location $\mathbf{x}_0$.
4. Predict the value $Z(\mathbf{x}_0)$. With the λ_i , the observed values $z(\mathbf{x}_i)$ and the estimated value of μ_Z in equation 5 the unknown value of $Z(\mathbf{x}_0)$ is predicted.

Spatial interpolation by kriging

Simple kriging: practical application

5. Calculate the **variance** of the **prediction error**. Using $\lambda_i(\mathbf{x}_0)$, $C_Z(\mathbf{h})$ and σ_Z^2 the variance of the prediction error is calculated with equation 11. The result is a map of predicted properties on a grid and a map of associated **error variances**. The figure below shows the map of kriging predictions and the associated **prediction variance** or **kriging variance**.

